

Freeside — Bachelor's Thesis Methodology Guide

Academic Research Design, Metrics & Literature Review

SECTION 1 — Defining & Measuring Productivity

How "Productivity" is Defined Across Disciplines

Productivity is one of the most cited yet most contested constructs in HCI and psychology research, precisely because it means something different in each field. Your thesis is stronger if you explicitly acknowledge this and choose a working definition that fits Freeside's scope.

In Industrial/Organizational Psychology

Productivity is traditionally defined as **output per unit of time** — a ratio model inherited from manufacturing. This is measurable in knowledge work as: tasks completed per hour, code commits per day, or words written per session. The problem is that it ignores quality and mental cost.

Landmark framing: Drucker, P.F. (1999). "Knowledge-Worker Productivity: The Biggest Challenge." California Management Review, 41(2), 79–94. Drucker argued that knowledge work productivity cannot be measured by output alone — it requires measuring goal alignment and self-direction. This paper directly justifies why Freeside's approach (energy-aligned task routing) is more valid than raw output counting.

In Cognitive Psychology

Productivity is framed as a function of **cognitive resources** — attention, working memory, and executive function. A person is "productive" when their cognitive resources are matched to the demands of the task (flow state). Overload → errors and burnout. Underload → boredom and avoidance.

Core framework: Sweller, J. (1988). "Cognitive load during problem solving: Effects on learning." Cognitive Science, 12(2), 257–285. This is your primary theoretical anchor. Cognitive Load Theory states that working memory has finite capacity. When task demands exceed this capacity, performance collapses. Freeside's routing algorithm is a direct software implementation of this theory.

In HCI and Personal Informatics

HCI researchers define productivity through behavioral signals rather than pure output. The most cited framework is the **Personal Informatics Model** by Li, Dey & Forlizzi, which describes how people collect and reflect on personal data to change behavior.

Core framework: Li, I., Dey, A., & Forlizzi, J. (2010). "A stage-based model of personal informatics systems." CHI '10 Proceedings. ACM. This is your methodological blueprint. Freeside is a personal informatics system: it collects energy and task data (collection stage), surfaces patterns (reflection stage), and adjusts behavior (action stage).

Established Methods for Measuring Personal Productivity

Method 1 — Self-Report Scales (Subjective)

The participant rates their own productivity, energy, or stress on a validated scale at set intervals.

Examples of validated instruments:

- **Perceived Productivity Scale (PPS)** — single-item or multi-item rating of how productive the participant felt during a period
- **PANAS (Positive and Negative Affect Schedule)** — Watson, Clark & Tellegen (1988) — measures positive/negative mood states that correlate with output
- **Procrastination Assessment Scale – Students (PASS)** — Solomon & Rothblum (1984) — measures frequency and reasons for academic procrastination

Pros: Fast to administer, captures subjective experience, low burden. **Cons:** Susceptible to recall bias, social desirability, and inconsistent self-awareness. People overestimate their own productivity (the "80% illusion" framing from your pitch deck is backed by research).

Method 2 — Objective Behavioral Tracking (Passive)

Log observable behaviors: number of tasks completed, time taken per task, task switching frequency, session length.

Examples:

- Task completion count per day
- Time-to-completion per cognitive load tier
- Number of AI co-pilot requests (indicator of friction/overwhelm)
- Days with completed morning energy check-ins (engagement proxy)

Pros: Objective, unbiased, collected automatically by the system. **Cons:** Does not capture quality of work, only quantity.
A fast completion could mean a poorly done task.

Method 3 — Experience Sampling Method (ESM)

Participants are prompted at random or fixed intervals throughout the day to report their current state: energy, focus, stress, task type.

Founding paper: Csikszentmihalyi, M. & Larson, R. (1987). "Validity and reliability of the experience-sampling method." Journal of Nervous and Mental Disease, 175(9), 526–536.

Pros: Captures in-the-moment data, avoids end-of-day recall bias. **Cons:** High participant burden, risk of dropout in longer studies. For your thesis: Freeside's morning energy check-in is a simplified, once-daily ESM protocol. Describe it as such.

Method 4 — Diary Studies

Participants keep a structured daily log of their work activities, energy levels, and obstacles.

Pros: Rich qualitative data, captures context machines cannot. **Cons:** High effort, variable quality, difficult to analyze at scale.

Recommended Approach for Your Thesis

Use a **mixed-methods design** combining:

- Objective behavioral data from Freeside's database (Method 2) — your primary metrics
- Pre/post validated questionnaires (Method 1) — baseline and outcome measurement
- Optional: brief end-of-day rating (3 questions, in-app) as a lightweight ESM

This combination is the standard approach in HCI intervention studies and gives you both quantitative rigor and subjective validity.

SECTION 2 — The 5 Core Metrics

How to Read This Section

For each metric you will find:

- **What it is** — definition and academic grounding
 - **Baseline measurement** — how to score it before using Freeside
 - **Post-intervention measurement** — how to score it after 2–4 weeks
 - **In-app data source** — what Freeside logs automatically
 - **Key papers** — where to find academic backing
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Metric 1 — Task Completion Rate (TCR)

What it is

The percentage of planned or intended tasks that are actually completed within a given time window. This is your most direct, quantitative measure of whether Freeside improves productive output.

Academic grounding: Goal-setting theory (Locke & Latham, 1990) establishes that specific, appropriately difficult goals increase performance. Your routing algorithm operationalizes this by surfacing tasks calibrated to current capacity.

Baseline Measurement (Pre-Freeside)

Administer a 1-week diary study before participants begin using Freeside. Ask them to log every day:

- How many tasks did you intend to complete today? (planned)
- How many did you actually complete? (actual)
- Rate your satisfaction with today's output: 1–5

Calculate: $TCR = (\text{tasks completed} / \text{tasks planned}) \times 100$

Typical baseline TCR in literature is around 50–65% for knowledge workers (Bellotti et al., 2004, CHI).

Post-Intervention Measurement

After 2–4 weeks of Freeside use, extract from the database:

- Tasks created per day (planned)

- Tasks marked complete per day (actual)
- Calculate the same TCR formula
- Compare using paired t-test or Wilcoxon signed-rank test

In-App Data Source

`session_logs.completed_at` vs `tasks.created_at` The ratio of completed to created gives you TCR automatically.

Scoring System for Thesis

TCR Score	Rating
< 40%	Critical — severely below capacity
40–60%	Poor — common baseline
60–75%	Moderate — average knowledge worker
75–90%	Good — Freeside target range
> 90%	Excellent

Key Papers

- Locke, E.A. & Latham, G.P. (1990). A theory of goal setting and task performance. Prentice Hall.
- Bellotti, V., Dalal, B., Good, N., et al. (2004). "What a to-do: Studies of task management towards the design of a personal task list manager." CHI '04, ACM. ← directly relevant, measures real-world task completion

Metric 2 — Cognitive Load Index (CLI)

What it is

A measure of the mental effort required to perform tasks relative to the performer's available cognitive resources. When task demand exceeds capacity, errors increase and speed decreases. Freeside's entire routing algorithm is a direct intervention on this metric.

Academic grounding: Sweller (1988) Cognitive Load Theory; Kahneman, D. (1973). "Attention and effort." Prentice Hall — defines cognitive resources as finite and allocatable.

Baseline Measurement

Use the validated **NASA Task Load Index (NASA-TLX)** questionnaire. It measures six subscales: Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration. Each rated 0–100. Calculate a weighted composite score.

Administer at end of each day during the 1-week pre-study period.

Download: NASA-TLX is public domain. Find it at: humansystems.arc.nasa.gov/groups/tlx/

Post-Intervention Measurement

Administer the same NASA-TLX weekly during Freeside use. Compare composite scores: pre-study average vs. Week 2 average vs. Week 4 average. Hypothesis: CLI decreases as Freeside routes appropriately-loaded tasks, reducing mismatch between capacity and demand.

In-App Data Source

Proxy measure: **rerouting rate** from `session_logs.was_rerouted`. A high rerouting rate on low-energy days = system correctly identifying overload conditions. Track reroute frequency over time — declining reroutes with stable completion rate suggests the user is internalizing better habits.

Scoring System

NASA-TLX composite score: | Score Range | Interpretation | |-----|-----| | 0–25 | Low load — underutilization | | 26–50 | Moderate — productive zone | | 51–75 | High — approaching overload | | 76–100 | Critical overload — burnout risk |

Key Papers

- Hart, S.G. & Staveland, L.E. (1988). "Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research." *Human Mental Workload*, 1(3), 139–183. ← the original NASA-TLX paper
- Sweller, J., van Merriënboer, J., & Paas, F. (1998). "Cognitive architecture and instructional design." *Educational Psychology Review*, 10(3), 251–296.

Metric 3 — Sleep Quality & Duration Score (SQDS)

What it is

Sleep is the single most evidence-backed predictor of next-day cognitive performance, emotional regulation, and decision quality. Including sleep as a metric in your thesis connects Freeside's well-being philosophy to the strongest physiological evidence base available.

Academic grounding:

- Van Dongen, H.P.A., et al. (2003): 17 consecutive nights of 6h sleep causes cognitive impairment equivalent to 48h total sleep deprivation — yet people don't notice the decline.
- Walker, M.J. & Stickgold, R. (2006): REM sleep specifically consolidates procedural and emotional memory, critical for creative and strategic work.

Baseline Measurement

Use the **Pittsburgh Sleep Quality Index (PSQI)** — the most widely used validated sleep questionnaire in clinical and research settings.

It covers 7 components: subjective sleep quality, sleep latency (how long to fall asleep), sleep duration, sleep efficiency, sleep disturbances, use of sleep medication, and daytime dysfunction.

Composite PSQI score: 0–21. Score > 5 = poor sleeper.

Administer at the start of the study (1-week baseline) and at end of the Freeside study period.

PSQI is free for research use. Find it at: sleep.pitt.edu/instruments/

How Freeside Connects to Sleep (Your Thesis Argument)

Freeside does not directly track sleep. But it affects the factors that destroy sleep quality:

- By preventing late-night task overload (routing heavy tasks to better windows)
- By reducing anxiety from unfinished task lists (the rerouting mechanism removes "overdue" guilt — which is a known sleep disruptor per Berset et al., 2011)
- By structuring work sessions with micro-steps that reduce end-of-day unfinished business (Zeigarnik effect)

This causal chain is your thesis argument for sleep as a metric. You are not claiming Freeside is a sleep app. You are claiming that better task management → lower work-related rumination → better sleep.

Post-Intervention Measurement

Repeat PSQI at end of study. Additionally, add a simple in-app daily question (optional, evening prompt): "How many hours did you sleep last night?" (1–12h) and "How rested did you wake up feeling?" (1–5).

This gives you a longitudinal sleep trajectory to plot alongside energy scores.

Scoring System

Use PSQI directly: | PSQI Score | Sleep Quality | |-----|-----| | 0–4 | Good sleeper | | 5–10 | Poor sleeper — intervention warranted | | 11–21 | Severely disturbed sleep |

Hypothesis: participants with PSQI > 5 at baseline show improvement after 2–4 weeks of Freeside use, mediated by reduced work-related evening stress.

Key Papers

- Buysse, D.J., Reynolds, C.F., Monk, T.H., et al. (1989). "The Pittsburgh Sleep Quality Index: A new instrument for psychiatric practice and research." *Psychiatry Research*, 28(2), 193–213. ← original PSQI paper, ~20,000 citations
 - Van Dongen, H.P.A., Maislin, G., Mullington, J.M., & Dinges, D.F. (2003). "The cumulative cost of additional wakefulness." *Sleep*, 26(2), 117–126.
 - Berset, M., Elfering, A., Lüthy, S., et al. (2011). "Work stressors and impaired sleep: Rumination as a mediator." *Stress and Health*, 27(2).
 - Pilcher, J.J. & Huffcutt, A.J. (1996). "Effects of sleep deprivation on performance: A meta-analysis." *Sleep*, 19(4), 318–326.
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Metric 4 — Procrastination Frequency Index (PFI)

What it is

Procrastination is not a time management problem — it is an emotion regulation problem. People avoid tasks not because they are lazy, but because the task triggers negative emotions (anxiety, boredom, self-doubt) that avoidance temporarily relieves. Freeside's micro-step generation and energy-matched routing are direct behavioral interventions on this mechanism.

Academic grounding:

- Steel, P. (2007): "The Nature of Procrastination: A Meta-Analytic and Theoretical Review of Quintessential Self-Regulatory Failure." *Psychological Bulletin*, 133(1), 65–94. — the definitive meta-analysis on procrastination. Steel introduces the **Temporal Motivation Theory (TMT)**: the motivation to do a task = (expectancy × value) / (impulsiveness × delay). Freeside increases expectancy (by breaking down large tasks) and reduces effective delay (by routing urgent-but-manageable tasks to the surface).

Baseline Measurement

Use the **Pure Procrastination Scale (PPS)** — Steel (2010), 12 items, validated, freely available in the 2010 paper's appendix.

Or the shorter **Irrational Procrastination Scale (IPS)** — Steel (2010), 9 items, Cronbach's $\alpha = 0.91$.

Rate on a 5-point Likert scale. Higher score = more procrastination.

Administer at the start of the study.

Post-Intervention Measurement

Repeat PPS/IPS at end of the study.

Additionally, use **behavioral proxy data from Freeside**:

- **Task initiation delay**: time between a task being created/surfaced and the user clicking on it (can be logged if you add a `viewed_at` timestamp to `session_logs`).
- **Co-pilot breakdown requests**: each time a user clicks "Break this down" on a task is a direct procrastination intervention event. Track frequency.

Scoring System

PPS (12-item, 1–5 scale): | PPS Score | Procrastination Level | |-----|-----| | 12–24 | Low — good self-regulation | | 25–36 | Moderate — some avoidance patterns | | 37–48 | High — significant interference | | 49–60 | Severe — chronic procrastinator |

Hypothesis: PPS scores decrease after Freeside use, particularly on the subscales measuring task avoidance and inability to start.

Key Papers

- Steel, P. (2007). "The nature of procrastination." *Psychological Bulletin*, 133(1), 65–94. ← ~3,000 citations, your foundational procrastination paper
- Steel, P. (2010). "Arousal, avoidant and decisional procrastinators." *Personality and Individual Differences*, 48(8), 926–934.
- Sirois, F. & Pychyl, T. (2013). "Procrastination and the priority of short-term mood regulation: Consequences for future self." *Social and Personality Psychology Compass*, 7(2), 115–127. ← establishes procrastination as emotion regulation, not laziness
- Gropel, P. & Steel, P. (2008). "A mega-trial investigation of goal setting, interest enhancement, and energy on procrastination." *Personality and Individual Differences*, 45(5), 406–411.

Metric 5 — Subjective Well-Being & Burnout Risk Score (SWBBS)

What it is

Burnout is not a sudden event — it is a gradual erosion of energy, engagement, and efficacy. Measuring well-being and burnout risk as a metric ties together Freeside's psychological claims: that adaptive task management protects mental health alongside productivity.

Academic grounding:

- Maslach, C. & Leiter, M.P. (2016): Burnout is defined by three dimensions: exhaustion, cynicism (depersonalization), and reduced efficacy.
- Bakker, A.B. & Demerouti, E. (2007): Job Demands-Resources model — burnout occurs when job demands chronically exceed available resources (energy). Freeside is a JD-R intervention: it modulates which demands are presented based on available resources (energy score).

Baseline Measurement

Use the **Maslach Burnout Inventory – General Survey (MBI-GS)** or the more accessible **Oldenburg Burnout Inventory (OLBI)** — 16 items, two subscales: Exhaustion and Disengagement. Free for academic use.

Additionally use the **Perceived Stress Scale (PSS-10)** — Cohen, S. (1983), 10 items, extremely widely used, free for academic use, ~40,000 citations.

PSS-10 measures perceived stress and sense of control over one's situation.

Post-Intervention Measurement

Repeat OLBI and PSS-10 at end of study. Your hypothesis: OLBI Exhaustion subscale and PSS-10 scores decrease after Freeside use, as the system prevents chronic task overload.

In-App Data Source (Burnout Proxy)

Plot the user's **rolling 7-day average energy score** from `energy_logs`. A declining trend over multiple days = early burnout signal. Include this chart in your thesis alongside OLBI scores — if the in-app energy trend and the OLBI exhaustion scores move together, you have evidence that Freeside's energy tracking is a valid behavioral burnout proxy.

This is a novel finding you can claim as your own.

Scoring System

PSS-10 (0–40 total): | PSS-10 Score | Stress Level | |-----|-----| | 0–13 | Low stress | | 14–26 | Moderate stress | | 27–40 | High perceived stress |

OLBI Exhaustion subscale (1–4 scale, 8 items): Higher = more exhaustion.

Key Papers

- Cohen, S., Kamarck, T., & Mermelstein, R. (1983). "A global measure of perceived stress." *Journal of Health and Social Behavior*, 24(4), 385–396. ← original PSS paper, ~40,000 citations
 - Maslach, C., Schaufeli, W.B., & Leiter, M.P. (2001). "Job burnout." *Annual Review of Psychology*, 52(1), 397–422.
 - Bakker, A.B. & Demerouti, E. (2007). "The job demands-resources model: State of the art." *Journal of Managerial Psychology*, 22(3), 309–328.
 - Demerouti, E., Bakker, A.B., Nachreiner, F., & Schaufeli, W.B. (2001). "The job demands-resources model of burnout." *Journal of Applied Psychology*, 86(3), 499–512.
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Summary Table — All 5 Metrics

#	Metric	Baseline Instrument	In-App Proxy	Hypothesis
1	Task Completion Rate	1-week diary	session_logs completion ratio	TCR increases post-Freeside
2	Cognitive Load Index	NASA-TLX	rerouting rate	NASA-TLX score decreases
3	Sleep Quality Score	PSQI	evening energy rating	PSQI improves for poor sleepers
4	Procrastination Frequency	PPS / IPS	task initiation delay	PPS score decreases
5	Well-Being / Burnout Risk	OLBI + PSS-10	rolling energy trend	PSS-10 and OLBI decrease

SECTION 3 — Main Research Goal & Methodology

Formal Statement of the Research Goal

"This research investigates whether a software system that operationalizes empirically validated findings from cognitive psychology and behavioral science — specifically Cognitive Load Theory, Self-Determination Theory, and Temporal Motivation Theory — can measurably improve personal productivity outcomes, reduce procrastination, and lower burnout risk in knowledge workers and students, as compared to their unaided baseline state."

Or, for a shorter abstract version:

"This study presents Freeside, an adaptive AI-driven productivity platform designed to translate established behavioral science into personalized, real-time task management interventions. Using a pre-test/post-test quasi-experimental design with N=10–15 participants over four weeks, we evaluate Freeside's impact on five validated productivity-related metrics: task completion rate, cognitive load, sleep quality, procrastination frequency, and subjective well-being."

Formal Methodology Name & Structure

Primary Design: Pre-Test / Post-Test Quasi-Experimental Design

This is the most appropriate design for your situation. It is called "quasi-experimental" because you do not have a true control group (random assignment with a group that uses no tool). Instead, each participant serves as their own control: their baseline IS the control condition.

Formal structure:

Phase 1 (Week 0)	– Baseline measurement - Administer: PSQI, NASA-TLX, PSS-10, OLBI, PPS - Conduct: 1-week behavioral diary
Phase 2 (Weeks 1-4)	– Intervention period - Participants use Freeside daily - System logs all behavioral data automatically - Optional: weekly in-app PSS-3 pulse check
Phase 3 (Week 5)	– Post-intervention measurement - Re-administer: PSQI, NASA-TLX, PSS-10, OLBI, PPS - Extract: all Freeside database metrics - Optional: brief semi-structured interview (15 min)
Phase 4	– Analysis - Paired t-tests or Wilcoxon signed-rank tests (depending on data normality – use Shapiro-Wilk test first) - Effect sizes: Cohen's d - Visualization: pre/post bar charts for each metric

Secondary Design Layer: Technology Probe Study

You should also formally describe Freeside as a **Technology Probe** (Hutchinson et al., 2003 — described below in literature). Technology probes are functional but not finished systems, deployed to real users to gather data about behavior and needs. This framing is common in HCI theses and explains why a Bachelor's MVP is academically sufficient — you are not claiming a finished product, you are using Freeside as a research instrument.

Why Not a Randomized Controlled Trial (RCT)?

An RCT would require 30+ participants per group, a control group using a different tool (or no tool), and random assignment. This is not feasible for a Bachelor's project timeline with 10–15 participants.

How to handle this in your thesis: Write in your Limitations section: "The absence of a randomized control group means observed improvements cannot be exclusively attributed to Freeside, as learning effects, seasonal variation, or regression to the mean may contribute. A future study with a larger sample and waitlist control group would strengthen causal claims."

This is the correct academic posture — honest, not defensive.

Statistical Analysis Plan

For each metric:

1. **Test normality:** Shapiro-Wilk test on pre and post scores
2. **If normal:** Paired samples t-test (compare means, report t, df, p, Cohen's d)
3. **If not normal:** Wilcoxon signed-rank test (report W, Z, p, r effect size)
4. **Significance threshold:** $\alpha = 0.05$ (standard)
5. **Effect size interpretation:** Cohen's d: small=0.2, medium=0.5, large=0.8

In Python:

```
from scipy import stats
import numpy as np

pre_scores = [...] # e.g., pre-study PSS-10 scores
post_scores = [...] # e.g., post-study PSS-10 scores

# Normality test
stat, p_normality = stats.shapiro(pre_scores)

# Paired t-test
t_stat, p_value = stats.ttest_rel(pre_scores, post_scores)

# Cohen's d
d = (np.mean(pre_scores) - np.mean(post_scores)) / np.std(pre_scores)

print(f"t={t_stat:.3f}, p={p_value:.3f}, Cohen's d={d:.3f}")
```

SECTION 4 — Literature Review: Keywords, Frameworks & Landmark Papers

4.1 — Databases to Search

- **ACM Digital Library** (dl.acm.org) → best for HCI papers
- **IEEE Xplore** (ieeexplore.ieee.org) → best for software systems papers
- **Google Scholar** → broad, good for citation tracking
- **PubMed** → for sleep and psychology papers
- **PsycINFO** (via your university library) → psychology validated instruments

4.2 — By Category

Cognitive Load & Attention

Search queries:

- "Cognitive load theory knowledge work"
- "Mental workload measurement productivity"
- "Attention management workplace HCI"
- "Task switching cognitive cost"

Landmark papers:

- Sweller, J. (1988). Cognitive Science. ← your foundational theory
 - Kahneman, D. (1973). Attention and Effort. Prentice Hall.
 - Mark, G., Gudith, D., & Klocke, U. (2008). "The cost of interrupted work: More speed and stress." CHI '08, ACM.
← interruptions cost 23 min recovery
 - Czerwinski, M., Horvitz, E., & Wilhite, S. (2004). "A diary study of task switching and interruptions." CHI '04, ACM.
-

Personal Productivity & Knowledge Work

Search queries:

- "Personal productivity measurement knowledge worker"
- "Task management system evaluation"
- "Self-regulation goal pursuit"
- "Personal informatics behavior change"

Landmark papers:

- Drucker, P.F. (1999). California Management Review.
 - Bellotti, V., et al. (2004). "What a to-do." CHI '04. ← task list study
 - Li, I., Dey, A., & Forlizzi, J. (2010). "A stage-based model of personal informatics systems." CHI '10. ← your PI framework
 - Rooksby, J., Rost, M., Morrison, A., & Chalmers, M. (2014). "Personal tracking as lived informatics." CHI '14. ← how people actually use tracking apps
-

Self-Determination Theory & Motivation

Search queries:

- "Self-determination theory technology motivation"
- "Intrinsic motivation goal setting software"
- "Autonomy competence relatedness productivity"

Landmark papers:

- Deci, E.L. & Ryan, R.M. (2000). "The 'what' and 'why' of goal pursuits." Psychological Inquiry, 11(4), 227–268. ← SDT foundational paper
 - Ryan, R.M. & Deci, E.L. (2017). Self-Determination Theory: Basic Psychological Needs in Motivation, Development, and Wellness. Guilford Press.
 - Przybylski, A.K., Rigby, C.S., & Ryan, R.M. (2010). "A motivational model of video game engagement." Review of General Psychology. ← SDT + gamification
-

Procrastination & Behavioral Activation

Search queries:

- "Procrastination self-regulation failure meta-analysis"
- "Temporal motivation theory delay"
- "Behavioral activation habit formation"
- "Task avoidance emotion regulation"

Landmark papers:

- Steel, P. (2007). Psychological Bulletin, 133(1). ← the procrastination bible
 - Sirois, F. & Pychyl, T. (2013). Social and Personality Psychology Compass.
 - Baumeister, R.F., & Tierney, J. (2011). Willpower: Rediscovering the Greatest Human Strength. Penguin. (popular, but cites academic research well)
 - Ferrari, J.R., Johnson, J.L., & McCown, W.G. (1995). Procrastination and Task Avoidance. Springer. ← academic depth
-

Sleep & Cognitive Performance

Search queries:

- "Sleep deprivation cognitive performance meta-analysis"
- "Sleep quality workplace productivity"
- "Work rumination sleep quality"
- "Pittsburgh Sleep Quality Index validation"

Landmark papers:

- Buysse, D.J., et al. (1989). Psychiatry Research. ← original PSQI
 - Van Dongen, H.P.A., et al. (2003). Sleep, 26(2). ← cumulative sleep debt
 - Pilcher, J.J. & Huffcutt, A.J. (1996). Sleep, 19(4). ← meta-analysis
 - Berset, M., et al. (2011). Stress and Health, 27(2). ← work stress → poor sleep
 - Åkerstedt, T., et al. (2002). "Work load and work hours in relation to disturbed sleep and fatigue." Journal of Psychosomatic Research, 53(3).
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Gamification & Behavioral Change

Search queries:

- "Gamification behavior change review"
- "Points badges leaderboards motivation"
- "Habit formation feedback loops HCI"
- "Behavioral design micro-habits"

Landmark papers:

- Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011). "From game design elements to gamefulness: Defining gamification." MindTrek '11, ACM. ← most cited gamification definition paper
 - Hamari, J., Koivisto, J., & Sarsa, H. (2014). "Does gamification work? A literature review." HICSS '14, IEEE.
 - Clear, J. (2018). Atomic Habits. Avery. (popular but well-grounded)
 - Fogg, B.J. (2009). "A behavior model for persuasive design." Persuasive Technology '09, ACM. ← Fogg Behavior Model: behavior = motivation + ability + trigger
-

HCI Methodology & Evaluation

Search queries:

- "Technology probe methodology HCI"
- "Pre-test post-test quasi-experimental HCI evaluation"
- "Experience sampling method productivity"
- "Within-subjects design software evaluation"

Landmark papers:

- Hutchinson, H., et al. (2003). "Technology probes: Inspiring design for and with families." CHI '03, ACM. ← technology probe methodology
 - Lazar, J., Feng, J.H., & Hochheiser, H. (2017). Research Methods in Human-Computer Interaction. Morgan Kaufmann. ← your methodology textbook
 - Consolvo, S. & Walker, M. (2003). "Using the experience sampling method to evaluate ubicomp applications." IEEE Pervasive Computing, 2(2).
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Burnout & Job Demands-Resources Model

Search queries:

- "Job demands resources model burnout"
- "Maslach burnout inventory validation"
- "Work engagement recovery resources"
- "Oldenburg burnout inventory psychometric"

Landmark papers:

- Maslach, C., Schaufeli, W.B., & Leiter, M.P. (2001). Annual Review of Psychology. ← burnout definition and MBI

- Bakker, A.B. & Demerouti, E. (2007). Journal of Managerial Psychology.
 - Demerouti, E., et al. (2001). Journal of Applied Psychology. ← JD-R model
 - Schaufeli, W.B. & Bakker, A.B. (2004). "Job demands, job resources, and their relationship with burnout and engagement." Journal of Organizational Behavior, 25(3).
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4.3 — How to Cite AI-Assisted Tools in Your Thesis

If your committee asks about your use of AI (Claude, etc.) in building Freeside, cite it transparently in your Methods section:

"The AI co-pilot feature of Freeside utilizes the Claude API (Anthropic, 2024) to deliver context-aware coaching. The system prompt is designed by the author to operationalize the behavioral principles of Self-Determination Theory and Cognitive Load Theory within each conversational interaction."

Putting It All Together — Your Thesis Structure Outline

Chapter 1 – Introduction

1.1 Problem statement (burnout, rigid tools)

1.2 Research questions:

- RQ1: Does energy-aligned task routing improve TCR?
- RQ2: Does Freeside reduce perceived cognitive load?
- RQ3: Is there a measurable relationship between Freeside use and sleep quality improvement?
- RQ4: Does Freeside reduce procrastination frequency?
- RQ5: Does Freeside lower burnout risk indicators?

1.3 Thesis structure

Chapter 2 – Literature Review

- 2.1 Productivity in HCI and psychology
- 2.2 Cognitive Load Theory (Sweller, 1988)
- 2.3 Self-Determination Theory (Deci & Ryan)
- 2.4 Temporal Motivation Theory & procrastination (Steel, 2007)
- 2.5 Sleep and cognitive performance
- 2.6 Gamification and behavioral change
- 2.7 Personal informatics systems (Li et al., 2010)

Chapter 3 – System Design (Freeside)

- 3.1 Design philosophy and psychological grounding
- 3.2 System architecture
- 3.3 Core features mapped to theory:
 - Energy routing → Cognitive Load Theory
 - Co-pilot → SDT (autonomy + competence)
 - Micro-steps → Procrastination intervention
 - Progress feedback → Behavioral activation
 - Calendar inference → JD-R model

Chapter 4 – Methodology

- 4.1 Research design (pre-test/post-test quasi-experimental)
- 4.2 Participants (N, recruitment, inclusion criteria)
- 4.3 Instruments (PSQI, NASA-TLX, PSS-10, OLBI, PPS)
- 4.4 Procedure (Phase 1-4)
- 4.5 In-app behavioral data collection
- 4.6 Statistical analysis plan
- 4.7 Ethical considerations (data privacy, Supabase RLS)

Chapter 5 – Results

- 5.1 Descriptive statistics (pre/post for each metric)
- 5.2 Inferential statistics (t-tests / Wilcoxon, effect sizes)
- 5.3 Behavioral data from Freeside database (4 charts)

5.4 Correlation analysis (e.g., energy score vs. TCR)

Chapter 6 – Discussion

6.1 Interpretation of findings per RQ

6.2 Theoretical implications

6.3 Practical implications for tool design

6.4 Limitations

Chapter 7 – Conclusion & Future Work

7.1 Summary of findings

7.2 LSTM burnout prediction (future)

7.3 Multi-platform integration (future)

7.4 Mobile application (future)

7.5 Larger RCT study (future)

References

Appendices

A: Survey instruments (PSQI, NASA-TLX, PSS-10, OLBI, PPS)

B: System screenshots

C: Raw statistical output

D: Participant consent form